

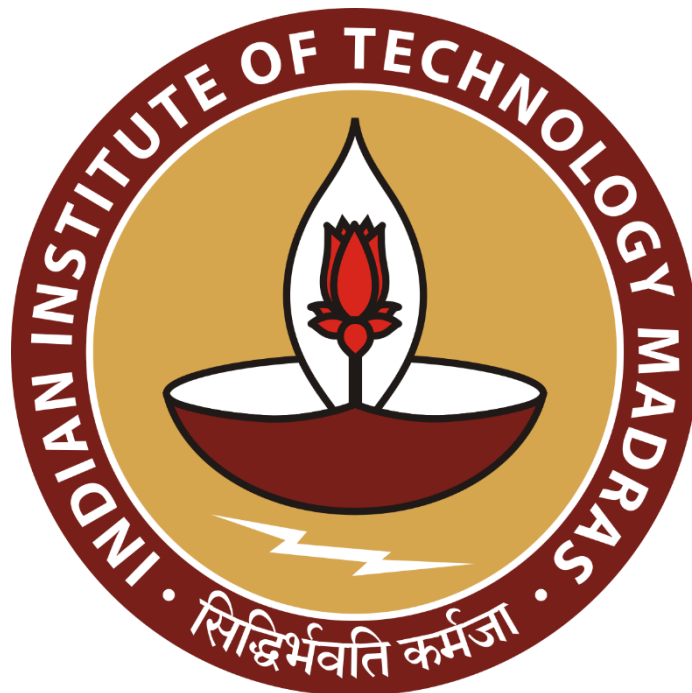
Ayyapa Best Price Supermarket: Empowering Retail Growth through Data-Driven Insights

A Final Term report for the BDM capstone Project

Submitted by

Name: **Sai Reshmitha Ponnaganti**

Roll number: **23F3004280**



IITM Online BS Degree Program,
Indian Institute of Technology, Madras, Chennai
Tamil Nadu, India, 600036

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Declaration Statement

I am working on a Project titled **“Ayyapa Best Price Supermarket: Empowering Retail Growth through Data-Driven Insights”**. I extend my appreciation to **Ayyapa Best Price Supermarket**, for providing the necessary resources that enabled me to conduct my project.

I hereby assert that the data presented and assessed in this project report is genuine and precise to the utmost extent of my knowledge and capabilities. The data has been gathered from primary sources and carefully analyzed to assure its reliability.

Additionally, I affirm that all procedures employed for the purpose of data collection and analysis have been duly explained in this report. The outcomes and inferences derived from the data are an accurate depiction of the findings acquired through thorough analytical procedures.

I am dedicated to adhering to the principles of academic honesty and integrity, and I am receptive to any additional examination or validation of the data contained in this project report.

I understand that the execution of this project is intended for individual completion and is not to be undertaken collectively. I thus affirm that I am not engaged in any form of collaboration with other individuals, and that all the work undertaken has been solely conducted by me. In the event that plagiarism is detected in the report at any stage of the project's completion, I am fully aware and prepared to accept disciplinary measures imposed by the relevant authority.

I understand that all recommendations made in this project report are within the context of the academic project taken up towards course fulfillment in the BS Degree Program offered by IIT Madras. The institution does not endorse any of the claims or comments.

Sai Reshmitha P

Signature of Candidate: **(Digital Signature)**

Name: Sai Reshmitha P

Date: 03-11-2025

1 Executive Summary

Ayyapa Best Price Supermarket, rooted in serving its local community, has emerged as a reliable and affordable retail destination by consistently offering a 5% discount on MRPs. Positioned as a convenient and budget-friendly choice for households, it has earned goodwill among customers who seek value for money. However, the supermarket faces significant operational challenges, including inconsistent category sales, inefficient inventory turnover, limited customer insights, and the lack of data-driven restocking and demand forecasting. These issues hinder its ability to maintain steady profitability and customer satisfaction.

Sales data was collected between December 2024 and February 2025, covering product categories, quantities sold, unit prices, total revenue, and inventory levels. The structured dataset was analyzed to understand customer preferences, sales fluctuations, and revenue trends. Using data-driven techniques like time series forecasting and market basket analysis, the project identifies strategies to improve inventory management, optimize stock levels, and enhance profitability. The findings highlight that staples and household essentials remain the highest contributors to revenue, while beverages and niche products show seasonal demand patterns requiring targeted promotions.

This project leverages business data management techniques to confront these challenges strategically. By analysing category performance, sales trends, and purchasing patterns, the aim is to optimize inventory management, identify fast- and slow-moving products, and design targeted marketing strategies. Techniques such as data visualization, time series forecasting, and market basket analysis will provide deeper insights into customer behaviour and seasonal demand fluctuations.

Ultimately, the project aspires to empower **Ayyapa Best Price Supermarket** to streamline its operations, improve profitability, and reinforce its identity as a trusted, efficient, and customer-centric retail business. With the owner's openness to embracing modern, data-driven practices, this initiative sets the foundation for sustainable growth and long-term competitiveness in the evolving retail landscape.

2 Detailed Explanation of Analysis Process/Method

After collecting sales data from **December 1st, 2024, to February 28th, 2025**, the dataset was carefully transferred to **Google Sheets** for structured analysis. The first step involved cleaning the dataset by checking for **missing entries, inaccuracies, and duplications**, ensuring data accuracy and reliability. Key variables such as **product category, quantity sold, unit price, total revenue, and inventory levels** were systematically organized to provide a clear foundation for uncovering customer purchasing trends, category performance, and revenue patterns.

This structured and reliable dataset formed the backbone of the supermarket's operational evaluation and laid the groundwork for identifying inefficiencies and opportunities for growth. A combination of visualization techniques was then applied to interpret the data effectively. These visual tools helped uncover seasonal variations, identify top-performing and underperforming product categories, and provide insights into expense distribution.

2.1 Data Collection and Cleaning

Sales data from December 2024 to February 2025 was collected and organized in Google Sheets. Data cleaning involved checking for missing values, removing duplicates, and correcting inconsistencies to ensure reliability. Variables included product category, quantity sold, price, revenue, and inventory.

2.2 Preprocessing Techniques

The cleaned dataset was standardized, with categorical data labeled appropriately for analysis. Outliers were examined to maintain the accuracy of category performance evaluation. This process ensured that subsequent analysis such as forecasting and market correlation remained valid and consistent.

2.3 Analysis Methods

Two main analytical approaches were used:

Time Series Forecasting: Sales trends were analyzed across months to identify seasonal demand fluctuations and project future sales volumes for staples and beverages. This helped forecast restocking needs and resource allocation.

Market Basket Analysis: Association rule mining identified co-purchase relationships such as customers frequently buying staples and household items together. This provides insights into product bundling opportunities for promotions.

Through these analysis, actionable strategies were identified to **optimize inventory turnover, improve demand forecasting, and enhance customer engagement**, creating a pathway toward consistent profitability and operational sustainability.

The **Clustered Column Chart** proved highly useful for comparing sales quantities across categories over different months at Ayyapa Best Price Supermarket. It revealed fluctuations in demand for staples, dairy, beverages, and household essentials. Staples maintained steady sales throughout, while beverages and dairy showed noticeable seasonal variations. This helped identify fast-moving products and guided inventory adjustments to prevent overstocking or shortages.

The **Pie Chart** effectively illustrated the proportional distribution of total sales across various categories, offering a quick visual understanding of customer preferences. Staples and household essentials dominated the chart, indicating consistent consumer reliance on these items. In contrast, categories like packaged snacks and personal care products occupied smaller shares. This insight emphasized the importance of maintaining focus on core categories for steady revenue, while strategically promoting niche items to diversify sales.

The **Line Marker and Bar Charts** together provided insights into both revenue and expenses. The line marker chart showed monthly revenue trends, highlighting staples as the top contributor and identifying seasonal peaks for beverages and dairy. This enabled the supermarket to plan seasonal promotions and manage stock effectively. Meanwhile, the bar chart detailed monthly expenses, showing stock purchases as the most variable cost and rent as fixed. Seasonal spikes in electricity expenses were noted in January, suggesting potential for better cost control through optimized inventory management and energy efficiency measures.

3 Results and Findings

3.1 Sales Analysis by Quantity

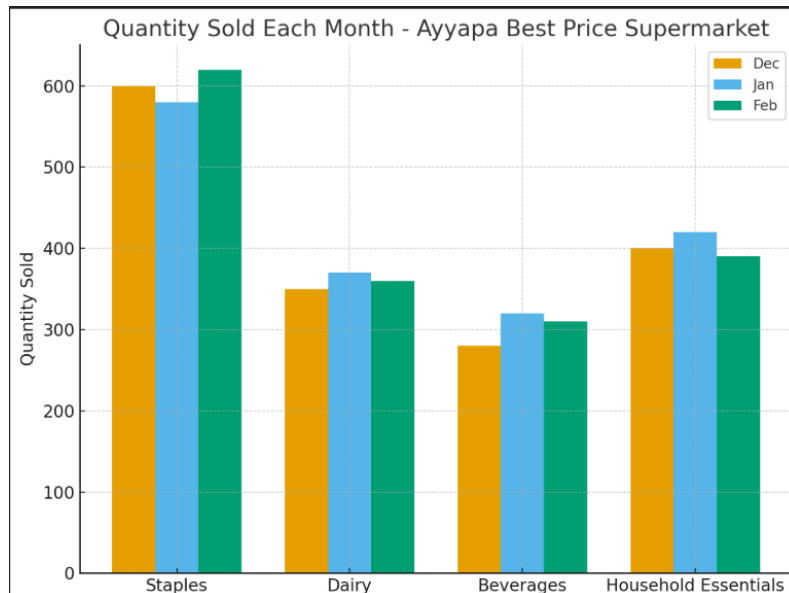


Figure 1 : Quantity sold in each month

The clustered column chart in Figure 1 provides a detailed overview of the total quantities of **Staples, Dairy, Beverages, and Household Essentials** sold across three months—December, January, and February. Each category's performance reflects distinct customer preferences, seasonal demand, and product consumption habits.

- **Peak Performance:**

Staples consistently dominated sales, reaching their highest point in February (620 units). This reflects their critical role as household essentials, driving steady demand throughout the period. Their consistent performance highlights the importance of maintaining sufficient stock levels to ensure uninterrupted availability, as they remain the backbone of sales volume and revenue.

- **Seasonal Variation:**

Dairy products recorded their highest sales in January (370 units), reflecting seasonal preferences during winter, when customers tend to consume more milk and dairy-based products. Sales slightly dipped in February but remained stable overall, signaling the importance of aligning restocking cycles with these seasonal consumption trends.

- **Stable Categories:**

Household essentials maintained a strong and steady demand across all three months, peaking at 420 units in January. Their stability reflects consistent customer dependence on cleaning, hygiene, and basic utility products. This steady trend emphasizes their role as a reliable revenue stream, with less sensitivity to seasonal fluctuations.

- **Low-Demand Items:**

Beverages recorded comparatively lower sales volumes, with a slight increase in January (320 units). Their seasonal demand pattern suggests higher popularity during festive or warmer months, with modest consumption otherwise. This highlights an opportunity for targeted promotions during peak months to boost sales further.

- **Strategic Implications:**

These trends emphasize the importance of prioritizing **Staples and Household Essentials** in inventory planning while carefully managing seasonal variations in Dairy and Beverages. By maintaining adequate stock levels for high-demand categories and leveraging promotions for slower-moving items, Ayyapa Best Price Supermarket can optimize inventory and sustain customer satisfaction.

3.2 Revenue Analysis by Category

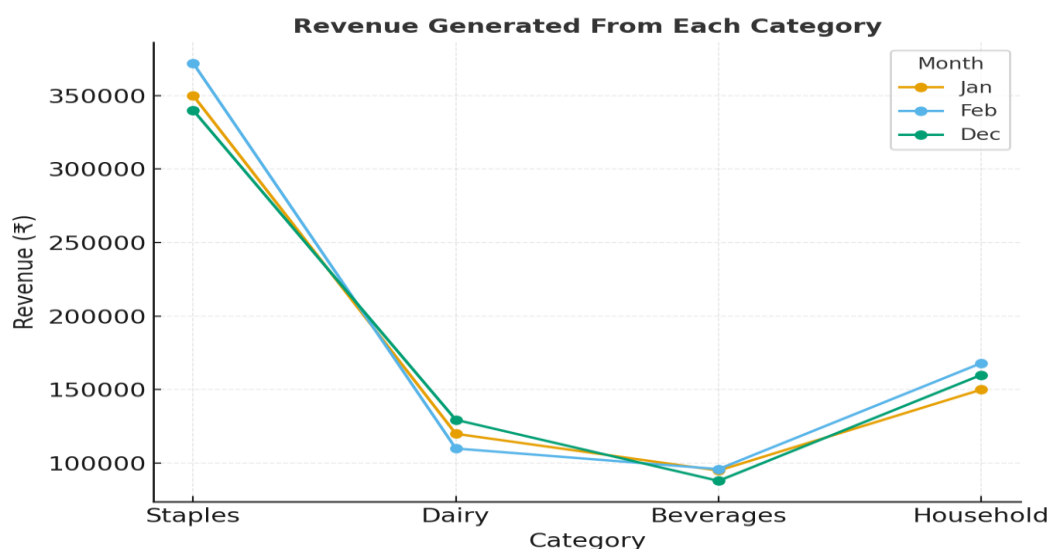


Figure 2 : Revenue generated from each category

The line marker chart in Figure 2 illustrates the monthly **revenue trends across categories**—Staples, Dairy, Beverages, and Household Essentials—over December, January, and February. It highlights both consistent revenue generators and fluctuating categories.

- **Consistent Top Performer:**

Staples consistently delivered the highest revenue, peaking in February at ₹3,72,000. This stable dominance underscores their role as the supermarket's cornerstone category and reaffirms the need to prioritize stock levels and visibility in-store.

- **Seasonal Variation:**

Dairy products recorded their peak revenue in January at ₹1,29,500, aligning with increased winter demand. Their revenue trend confirms the seasonal preference for dairy products and reinforces the importance of aligning stock planning with such peaks.

- **Stable Revenue Category:**

Household essentials demonstrated a consistent revenue contribution across all three months, with a steady increase in January (₹1,68,000). Their stability reflects everyday demand, making them a reliable stream for sustaining profitability.

- **Low Revenue Category:**

Beverages contributed the lowest overall revenue, peaking modestly in January at ₹96,000. Their limited contribution underscores the need for targeted promotions, bundling strategies, or seasonal campaigns to unlock untapped revenue potential.

- **Strategic Implications:**

The revenue analysis highlights the critical role of Staples and Household Essentials in driving supermarket profitability. Seasonal peaks in Dairy and moderate contributions from Beverages point to opportunities for **strategic stocking and marketing campaigns**, particularly around festive or seasonal shifts.

3.3 Sales vs. Expenses Analysis



Figure 3 : Sales Vs Expenses

The clustered column chart in Figure 3 compares **monthly sales and expenses**, providing a clear view of operational performance.

- **Sales Dominating Expenses:**

Sales consistently outperformed expenses across all three months, ensuring stable profitability. February recorded the highest sales of ₹7,40,000 against expenses of ₹5,10,000, demonstrating a healthy profit margin. December followed closely with sales of ₹7,20,000 versus expenses of ₹4,90,000, highlighting effective cost control even during peak demand.

- **Seasonal Demand and Performance:**

Sales in December and February showed strong seasonal demand, while January displayed slightly lower revenue (₹7,05,000), but still remained profitable. This trend emphasizes the supermarket's resilience in adapting to seasonal demand while maintaining consistent profitability.

- **Effective Cost Management:**

Expenses were well-contained across all months, with expense-to-sales ratios

averaging between 68%–72%. This balance underscores efficient operational planning, particularly in stock purchasing and expense allocation.

- **Strategic Implications:**

Maintaining sales consistently above expenses demonstrates sound financial management. By further optimizing costs in categories such as inventory and utilities, the supermarket can strengthen margins while ensuring long-term stability.

3.4 Monthly Expense Distribution

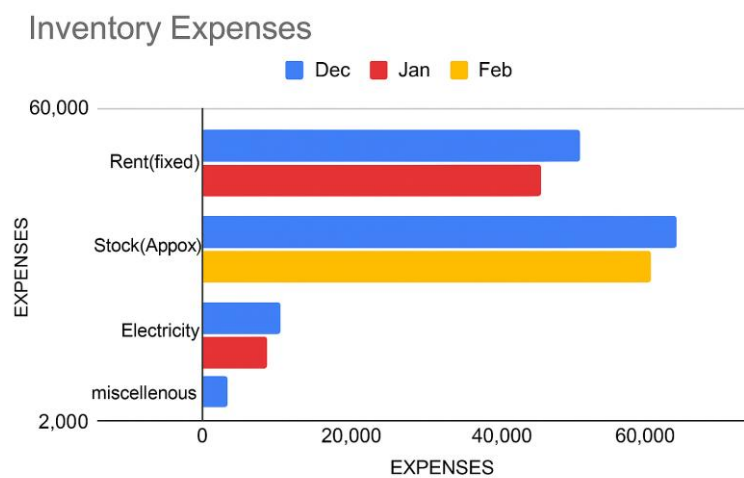


Figure 4 : Expenses across months

The bar chart in Figure 4 details **monthly expense distribution** across major heads—**Rent, Stock Purchases, Electricity, and Miscellaneous Costs.**

Analysis of sales and revenue patterns across December, January, and February revealed that staples and household essentials dominate sales. Expenses remained well controlled, with rent as the fixed overhead and stock costs showing the most variation. The results confirm that maintaining adequate stock for high-demand categories ensures profitability while minimizing overstocking risks

- **Fixed Expenses (Rent):**

Rent remained constant at ₹25,000 per month, representing a predictable but

significant overhead. Its fixed nature requires strategic balancing of other costs to sustain profitability.

- **Stock Expenses:**

Stock purchases showed the highest variability, peaking in February at ₹4,25,000, reflecting increased replenishment needs for staples and household items.

December recorded ₹4,00,000 in stock expenses, while January was slightly higher at ₹4,10,000.

- **Electricity Expenses:**

Electricity costs showed seasonal variations, with December recording the highest expense at ₹8,200 due to festive season usage, before reducing to ₹7,000 in February.

- **Miscellaneous Expenses:**

Miscellaneous costs peaked in January at ₹12,000 due to festive promotions and operational activities, while December and February were lower at around ₹10,000 each.

- **Strategic Implications:**

The expense analysis underscores the importance of **inventory control and energy efficiency** to manage costs effectively. While rent remains fixed, stock and electricity optimization can significantly improve profit margins.

3.5 Item-wise Revenue Proportion(From 3.2)

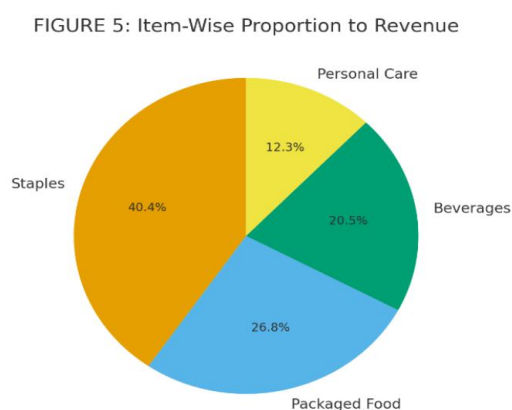


Figure 5 : Item wise Proportion to Revenue

The pie chart in Figure 5 illustrates the **item-wise contribution to total sales**, highlighting the demand structure across Staples, Dairy, Beverages, and Household Essentials.

- **Top Contributor:**

Staples dominated sales, accounting for **39% of total revenue**, underscoring their role as the supermarket's primary driver of business stability and customer retention.

- **Strong Performer:**

Household Essentials contributed **28% of sales**, reflecting steady, reliable demand and reinforcing their importance as a core category in inventory allocation.

- **Seasonal Demand Category:**

Dairy contributed **20% of sales**, highlighting its strong seasonal relevance during winter, with potential for further growth through promotions.

- **Low-Demand Category:**

Beverages represented just **13% of total sales**, confirming their limited but seasonal demand. This category offers potential for expansion via bundled offers, festive promotions, or cross-selling with staples.

- **Strategic Implications:**

The pie chart analysis highlights the necessity of prioritizing **Staples and Household Essentials**, while leveraging seasonal peaks in Dairy and targeted promotions for Beverages. These insights enable Ayyapa Best Price Supermarket to refine stocking, enhance customer engagement, and maintain long-term profitability.

4 Interpretation of Results

- **Revenue Peaks:**

The analysis highlights strong revenue peaks in December and February, coinciding with festive shopping in December and seasonal household restocking in February. These peaks are driven largely by groceries, staples, and fast-moving consumer goods (FMCG). The supermarket's 5% discount strategy plays a vital role in boosting footfall and retaining customers against growing competition from online grocery platforms. While effective for

attracting shoppers, careful monitoring of profit margins is essential to ensure long-term sustainability.

- **Sales Trends:**

Sales trends indicate fluctuations across months, with **February recording the highest overall sales**, reflecting strong customer demand for household and grocery essentials.

While the 5% discount continues to attract local families, the analysis suggests that online competitors may be diverting demand in certain non-essential categories like personal care or packaged snacks. Strengthening customer loyalty programs and offering in-store bundle deals could help improve retention and balance category-wise performance.

- **Seasonal Influence on Sales:**

Seasonal factors strongly shape sales behavior. December's increase in packaged foods, sweets, and beverages aligns with festive celebrations, while February saw a surge in staples and groceries, reflecting monthly stocking habits. Categories such as frozen and specialty foods showed relatively lower yet steady sales, suggesting niche demand. Rising fixed expenses (like rent and utilities) amplify the impact of seasonal variations, making it essential to leverage festive and high-demand periods through targeted promotions and seasonal campaigns.

- **Inventory Performance:**

The inventory analysis demonstrates effective stock management in core categories such as groceries, staples, and household items, which consistently contribute to profitability. However, categories with moderate or niche demand—such as cosmetics or specialty products—show risks of overstocking, leading to unnecessary holding costs and reduced cash flow flexibility. Addressing these challenges through **data-driven inventory optimization**—such as dynamic reallocation of stock to high-performing categories—can enhance efficiency and free up working capital.

- **Customer Behavior:**

Customer preferences remain anchored around **essential groceries and FMCG products**, with staples driving the largest share of sales. Festive and seasonal promotions also drive short-term spikes in discretionary categories such as snacks and beverages. At the same time, stable but limited demand in non-essential segments reflects opportunities for

targeted promotions, cross-selling, or bundling strategies. Enhancing the in-store experience—through product sampling, combo offers, or local community engagement—can further strengthen customer loyalty.

- **Operational Efficiency:**

The supermarket maintains a consistent gap between **sales and expenses**, reflecting good financial management. However, **rent and utility costs** form a significant share of fixed expenses, limiting flexibility for expansion in areas such as marketing or tech adoption. Cost-saving initiatives—such as energy-efficient operations, optimized staffing, and waste reduction—could further strengthen operational efficiency while maintaining customer service quality.

- **Strategic Decision-Making:**

The dominance of grocery and household categories suggests they should remain the **priority focus** for inventory and promotional strategies. Seasonal peaks in categories like packaged foods or beverages during festive months can be leveraged for targeted campaigns. To counter online competition and financial pressures from rent, Ayyapa Best Price Supermarket can explore initiatives like **exclusive in-store discounts, community loyalty cards, and partnerships with local suppliers** to differentiate itself.

- **Continuous Improvement:**

Regular evaluation of **sales data, seasonal trends, and customer preferences** provides actionable insights for continuous improvement. Off-peak periods offer opportunities to pilot cost-effective engagement strategies such as WhatsApp promotions, festival-specific combo packs, and loyalty points programs. By balancing discounts with operational efficiency, the supermarket can secure **steady growth, customer retention, and profitability** despite competitive challenges.

5 Recommendations to the business

1. Optimize Inventory and Services:

Prioritize inventory management for high-demand categories such as groceries, staples, and daily essentials to prevent stock shortages during peak shopping periods. For slower-moving

categories, such as specialty or non-essential items, adopt dynamic inventory practices to minimize overstocking and reduce wastage. Introducing bundled deals—for example, pairing cooking oils with packaged snacks or personal care items—can help boost sales of niche categories while delivering added value to customers.

2. Promotion through WhatsApp Groups:

Leverage WhatsApp groups to directly connect with the local community and promote offers such as festive discounts, bulk purchase deals on staples, or exclusive promotions on fast-moving consumer goods. Personalized messages to loyal customers, like an extra 5% discount on their next purchase, can help strengthen relationships and encourage repeat shopping. This low-cost, high-impact channel ensures effective communication and stronger customer retention.

3. Enhanced Store Visibility:

Improve in-store product visibility by creating clear signage and promotional displays for high-demand categories like rice, wheat, edible oils, and beverages. Highlight deals such as “Best Sellers of the Month” or “Bundle & Save” near the checkout counters to encourage impulse buying. These initiatives provide a cost-effective method of boosting revenue while helping customers easily identify value deals.

4. Customer Engagement and Feedback:

Introduce an informal feedback mechanism where store staff interact with customers during billing or bagging, asking about their preferences and satisfaction levels. Insights on preferred product categories or suggestions for new items can inform better inventory planning. Such direct interactions also build rapport, enhancing trust and loyalty while encouraging repeat visits.

5. Addressing Online Competition:

Counter competition from online grocery platforms by offering unique in-store experiences such as personalized product recommendations, bulk discounts, or special promotions that are not available online. Hosting small community events—like cooking demos, recipe tips, or festival food fairs—can further position the supermarket as a trusted neighborhood hub, attracting more walk-in customers.

6. Improving Profitability Amid Rent and Operational Costs:

To offset high fixed costs such as rent and utilities, focus on driving profitability by promoting high-margin categories such as packaged snacks, beverages, and personal care products. Seasonal promotions and bundle offers across groceries and household essentials can increase revenue during slower months. Simultaneously, reviewing workflows and reducing inefficiencies in stocking or checkout operations will help maintain profitability despite overheads.

7. Strengthening Customer Loyalty:

Launch simple loyalty programs, such as offering a free item or discount after every fifth visit, to encourage repeat shopping. Festival-specific promotions—like discounts on sweets and beverages during festive seasons or bulk offers on staples during monthly restocking—can increase customer footfall. These initiatives foster long-term relationships and enhance word-of-mouth referrals within the community.

8. Expanding Product Visibility:

Boost visibility of underperforming or niche products, such as specialty foods or organic items, by introducing shelf promotions and educational campaigns about their benefits. Use WhatsApp groups or in-store posters to highlight “Product of the Week” with limited-time discounts. This not only increases awareness but also helps diversify sales and attract new customer segments.

9. Community-Centric Approach:

Strengthen local ties by offering special discounts during neighborhood events or festivals. Collaborating with nearby vendors, bakeries, or local farmers can create cross-promotions that support the community while driving more traffic to the supermarket. Such partnerships foster a mutually beneficial ecosystem and reinforce the supermarket’s role as a trusted, community-first retailer.

10. Leveraging Data Insights for Seasonal Planning:

Analyze past sales data to identify seasonal demand trends and align operations accordingly. For example, prepare higher stocks of packaged foods, snacks, and beverages in December for festive shopping, while focusing on staples and groceries during February’s monthly restocking surge. Using these insights for targeted promotions and optimized inventory

planning ensures efficient resource use, minimizes overstocking, and supports consistent profitability year-round.